



Computing Infrastructures
September 11, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

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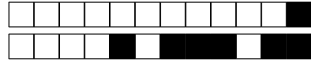
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Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the *Open Questions* should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.

**True false questions**

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 RAID 6 uses two parity stripes to provide redundancy.

☐ A False

☐ B True

Question 2 Open loop (free) cooling systems are less suitable for large-scale high-density datacenters where heat dissipation is a significant challenge.

☐ A False

☐ B True

Question 3 Paravirtualization requires modifications to the guest operating system to run on the virtual machine.

☐ A False

☐ B True

Question 4 TOR switches are also used for connecting servers within a rack to the datacenter network.

☐ A False

☐ B True

Question 5 Latency considerations have no relation on the global distribution of data centers.

☐ A True

☐ B False

Question 6 Software-as-a-Service (SaaS) cloud architectures provide users with solutions to develop and deploy their own applications.

☐ A True

☐ B False

Question 7 A two-tier leaf-spine topology without oversubscription (Clos topology) adopting routers with 6 ports can connect up to 18 servers .

☐ A True

☐ B False

Question 8 HDD capacity is typically subdivided into tracks and sectors, with files mapped to logical blocks.

☐ A False

☐ B True

Question 9 Cooling towers are a mandatory component in water-based cooling systems for datacenters.

☐ A True

☐ B False

Question 10 Edge computing always requires central data centers to process data.

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider a hard disk (HDD) with the following characteristics: block size of 2KB, mean seek time of 3ms, transfer time per block of 0.6ms, rotational speed of 12000 RPM, and negligible controller overhead. Knowing that the access locality is 84%, what is the average I/O time to read an entire file of 40MB?

Question 12

A datacenter operates a multi-tier web application composed of the following components both needed to process a request. Component A is the front-end layer composed of the load balancer (A1) and the front-end server (A2) required, both required for all incoming traffic. If A fails, no request can be processed. Component B is the application layer, composed of three identical and independent compute nodes (B1, B2, B3). Component B continues to operate as long as at least one compute node is available. We know that the availability of the load balancer and of the front-end server is 0.94 in both cases (A1 and A2). The availability of each compute node of the application layer (B1, B2, B3) is 0.7. Compute the overall system availability.



Question 13

We have to design a RAID 1+0 storage architecture composed of an array of 8 disks. Knowing that each disk has a MTTF equal to 450days and that we would like to have a MTTF for the storage infrastructure ($MTTF_{RAID1+0}$) higher than 8 years, what is the maximum MTTR that we have to consider to satisfy the requirement? Consider all the disks with the same characteristics and a single mirror case for the RAID 1 part.

Question 14

An enterprise application is deployed on a Web server, an Application server, and a Database server. The system is used by $N = 30$ users with a think time $Z = 20s$. During a monitoring interval of $T = 1200s$, you collect the following data:

- Number of completions: $C = 800$
- Web server utilization: $U = 50\%$
- Application server utilization: $U = 40\%$
- Database server utilization: $U = 60\%$

Compute the measured system throughput and service demands for all servers (i.e., D_{Web} , D_{App} , D_{DBMS})?



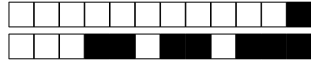
Question 15

Determine the maximum throughput and the minimum response time the system can achieve.

Question 16

You want to improve the overall system performance but have the budget to upgrade (replace with a fastest one) only a single server

- At which system layer should the upgrade be applied to achieve the greatest performance gain in terms of maximum achievable system throughput (i.e, *Web*, *App*, *DBMS*)?
- If the chosen server is upgraded so that its speed doubles compared to the initial configuration, what will be the new maximum achievable system throughput?



+1/6/55+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

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Question 17

⇒ Why virtualization is so important in modern datacenters?

Question 18

⇒ Explain how edge computing technologies integrate with traditional datacenters. What are the advantages and disadvantages of this hybrid architecture?

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.

[illegible]





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Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

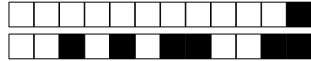
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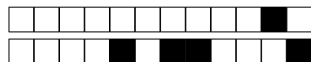
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Question 6 A two-tier leaf-spine topology without oversubscription (Clos topology) adopting routers with 6 ports can connect up to 18 servers .

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Question 7 HDD capacity is typically subdivided into tracks and sectors, with files mapped to logical blocks.

☐ A False

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Question 8 TOR switches are also used for connecting servers within a rack to the datacenter network.

☐ A False

☐ B True

Question 9 Cooling towers are a mandatory component in water-based cooling systems for datacenters.

☐ A False

☐ B True

Question 10 Latency considerations have no relation on the global distribution of data centers.

☐ A False

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Exercises

Correct answer: +2, No answer: 0.

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Consider a hard disk (HDD) with the following characteristics: block size of 2KB, mean seek time of 3ms, transfer time per block of 0.6ms, rotational speed of 10000 RPM, and negligible controller overhead. Knowing that the access locality is 87%, what is the average I/O time to read an entire file of 50MB?

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Question 13

We have to design a RAID 1+0 storage architecture composed of an array of 6 disks. Knowing that each disk has a MTTF equal to 400days and that we would like to have a MTTF for the storage infrastructure ($MTTF_{RAID1+0}$) higher than 12 years, what is the maximum MTTR that we have to consider to satisfy the requirement? Consider all the disks with the same characteristics and a single mirror case for the RAID 1 part.

Question 14

An enterprise application is deployed on a Web server, an Application server, and a Database server. The system is used by $N = 30$ users with a think time $Z = 20s$. During a monitoring interval of $T = 1200s$, you collect the following data:

- Number of completions: $C = 600$
- Web server utilization: $U = 50\%$
- Application server utilization: $U = 40\%$
- Database server utilization: $U = 60\%$

Compute the measured system throughput and service demands for all servers (i.e., D_{Web} , D_{App} , D_{DBMS})?



+2/5/46+

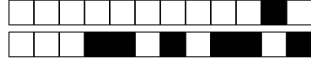
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Question 16

You want to improve the overall system performance but have the budget to upgrade (replace with a fastest one) only a single server

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+2/6/45+

Open Questions

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Question 17

⇒ Why virtualization is so important in modern datacenters?

Question 18

⇒ Explain how edge computing technologies integrate with traditional datacenters. What are the advantages and disadvantages of this hybrid architecture?

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True/False Questions

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Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

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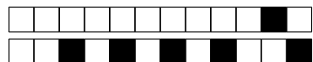
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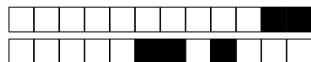
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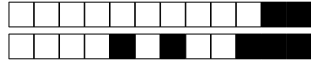
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☐ A False

☐ B True

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☐ B False

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Question 5 Open loop (free) cooling systems are less suitable for large-scale high-density datacenters where heat dissipation is a significant challenge.

☐ A True

☐ B False

Question 6 Edge computing always requires central data centers to process data.

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Question 7 Latency considerations have no relation on the global distribution of data centers.

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Question 9 TOR switches are also used for connecting servers within a rack to the datacenter network.

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☐ B True

Question 10 Cooling towers are a mandatory component in water-based cooling systems for datacenters.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

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Question 14

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+3/5/36+

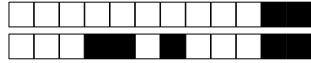
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+3/6/35+

Open Questions

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Answer Sheets (Page 3)

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True/False Questions

Question 01 : ☐A ☐B

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Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

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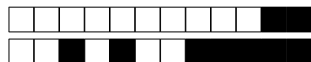
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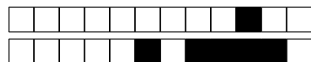
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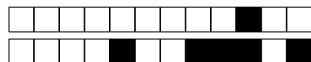
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☐ A True

☐ B False

Question 4 Edge computing always requires central data centers to process data.

☐ A True

☐ B False

Question 5 A two-tier leaf-spine topology without oversubscription (Clos topology) adopting routers with 6 ports can connect up to 18 servers .

☐ A False

☐ B True

Question 6 HDD capacity is typically subdivided into tracks and sectors, with files mapped to logical blocks.

☐ A True

☐ B False

Question 7 RAID 6 uses two parity stripes to provide redundancy.

☐ A False

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Question 8 Software-as-a-Service (SaaS) cloud architectures provide users with solutions to develop and deploy their own applications.

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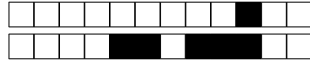
☐ A True

☐ B False

Question 10 TOR switches are also used for connecting servers within a rack to the datacenter network.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

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Question 11

Consider a hard disk (HDD) with the following characteristics: block size of 2KB, mean seek time of 3ms, transfer time per block of 0.6ms, rotational speed of 12000 RPM, and negligible controller overhead. Knowing that the access locality is 84%, what is the average I/O time to read an entire file of 40MB?

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A datacenter operates a multi-tier web application composed of the following components both needed to process a request. Component A is the front-end layer composed of the load balancer (A1) and the front-end server (A2) required, both required for all incoming traffic. If A fails, no request can be processed. Component B is the application layer, composed of three identical and independent compute nodes (B1, B2, B3). Component B continues to operate as long as at least one compute node is available. We know that the availability of the load balancer and of the front-end server is 0.96 in both cases (A1 and A2). The availability of each compute node of the application layer (B1, B2, B3) is 0.65. Compute the overall system availability.



Question 13

We have to design a RAID 1+0 storage architecture composed of an array of 8 disks. Knowing that each disk has a MTTF equal to 450days and that we would like to have a MTTF for the storage infrastructure ($MTTF_{RAID1+0}$) higher than 8 years, what is the maximum MTTR that we have to consider to satisfy the requirement? Consider all the disks with the same characteristics and a single mirror case for the RAID 1 part.

Question 14

An enterprise application is deployed on a Web server, an Application server, and a Database server. The system is used by $N = 30$ users with a think time $Z = 20s$. During a monitoring interval of $T = 1200s$, you collect the following data:

- Number of completions: $C = 600$
- Web server utilization: $U = 50\%$
- Application server utilization: $U = 40\%$
- Database server utilization: $U = 60\%$

Compute the measured system throughput and service demands for all servers (i.e., D_{Web} , D_{App} , D_{DBMS})?



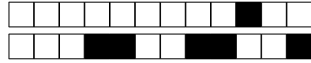
Question 15

Determine the maximum throughput and the minimum response time the system can achieve.

Question 16

You want to improve the overall system performance but have the budget to upgrade (replace with a fastest one) only a single server

- At which system layer should the upgrade be applied to achieve the greatest performance gain in terms of maximum achievable system throughput (i.e, *Web*, *App*, *DBMS*)?
- If the chosen server is upgraded so that its speed doubles compared to the initial configuration, what will be the new maximum achievable system throughput?



+4/6/25+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

⇒ Why virtualization is so important in modern datacenters?

Question 18

⇒ Explain how edge computing technologies integrate with traditional datacenters. What are the advantages and disadvantages of this hybrid architecture?

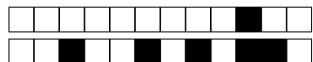
!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.

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Computing Infrastructures - September 11, 2025
Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

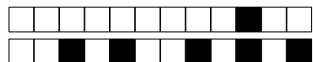
Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



+4/10/21+

Computing Infrastructures
September 11, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

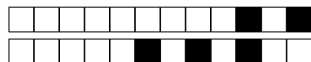
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

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If needed, you can use this page for notes. Any answer written here will be ignored.



+5/1/20+

Computing Infrastructures

September 11, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

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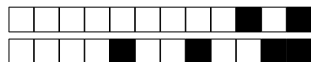
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 TOR switches are also used for connecting servers within a rack to the datacenter network.

☐ A True

☐ B False

Question 2 Latency considerations have no relation on the global distribution of data centers.

☐ A True

☐ B False

Question 3 Cooling towers are a mandatory component in water-based cooling systems for datacenters.

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☐ B False

Question 5 HDD capacity is typically subdivided into tracks and sectors, with files mapped to logical blocks.

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☐ B False

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Question 9 Software-as-a-Service (SaaS) cloud architectures provide users with solutions to develop and deploy their own applications.

☐ A True

☐ B False

Question 10 Open loop (free) cooling systems are less suitable for large-scale high-density datacenters where heat dissipation is a significant challenge.

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

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Question 14

An enterprise application is deployed on a Web server, an Application server, and a Database server. The system is used by $N = 30$ users with a think time $Z = 25s$. During a monitoring interval of $T = 1200s$, you collect the following data:

- Number of completions: $C = 600$
- Web server utilization: $U = 50\%$
- Application server utilization: $U = 40\%$
- Database server utilization: $U = 60\%$

Compute the measured system throughput and service demands for all servers (i.e., D_{Web} , D_{App} , D_{DBMS})?



+5/5/16+

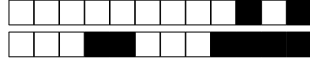
Question 15

Determine the maximum throughput and the minimum response time the system can achieve.

Question 16

You want to improve the overall system performance but have the budget to upgrade (replace with a fastest one) only a single server

- At which system layer should the upgrade be applied to achieve the greatest performance gain in terms of maximum achievable system throughput (i.e, *Web*, *App*, *DBMS*) ?
- If the chosen server is upgraded so that its speed doubles compared to the initial configuration, what will be the new maximum achievable system throughput?



+5/6/15+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

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Question 17

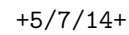
⇒ Why virtualization is so important in modern datacenters?

Question 18

⇒ Explain how edge computing technologies integrate with traditional datacenters. What are the advantages and disadvantages of this hybrid architecture?

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Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ Why virtualization is so important in modern datacenters?



●



Computing Infrastructures - September 11, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



Computing Infrastructures
September 11, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
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If needed, you can use this page for notes. Any answer written here will be ignored.



+6/1/10+

Computing Infrastructures September 11, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
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Question 1 TOR switches are also used for connecting servers within a rack to the datacenter network.

☐ A True

☐ B False

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Exercises

Correct answer: +2, No answer: 0.

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+6/4/7+

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- Application server utilization: $U = 40\%$
- Database server utilization: $U = 60\%$

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+6/5/6+

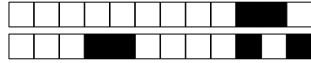
Question 15

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+6/6/5+

Open Questions

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Question 17

⇒ Why virtualization is so important in modern datacenters?

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⇒ Explain how edge computing technologies integrate with traditional datacenters. What are the advantages and disadvantages of this hybrid architecture?

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●



Computing Infrastructures - September 11, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

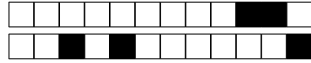
Question 12 :

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Question 14 :

Question 15 :

Question 16 :



+6/10/1+

Computing Infrastructures
September 11, 2025

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